

Guide to Working with Apache Airflow: Corrected Step-by-Step Setup

Follow these steps to set up Apache Airflow and run your first DAG:

Phase 1: Setup and Access

1. **Access the Environment:** Navigate to the following link and log in:
<https://killercoda.com/playgrounds/scenario/ubuntu>
2. **Start Airflow container:** Execute the command below.
`docker run -d -p 8080:8080 --name airflow apache/airflow standalone`
This starts Apache Airflow in the background.
3. **Check container status:** Execute: `docker ps`
*You should see a container named **airflow**.*
4. **Retrieve Admin Credentials:** Execute: `docker logs airflow`
*Scroll up through the output to locate and record the **admin** password.*
5. **Open Airflow UI:** In the environment, click the hamburger menu icon (top-right corner) and select 'traffic/ports.'
Log in using 'admin' as the username and the password you retrieved.

Phase 2: Create and Run the DAG

6. **Go inside the Airflow container:** Execute: `docker exec -it airflow bash`
7. **Move to DAGs folder:** Execute: `cd /opt/airflow/dags`
8. **Create DAG file:** Execute: `cat > first_dag.py`
Paste the following DAG code:

Python

```
from airflow import DAG
from airflow.operators.python import PythonOperator
from datetime import datetime

def task1_function():
    print('Task 1 Started')

def task2_function():
    print('Task 2 Running')

def task3_function():
    print('Task 3 Completed')
```

```

with DAG(
    dag_id='first_training_dag',
    start_date=datetime(2025, 1, 1),
    schedule='@daily',
    catchup=False,
    tags=['training']
) as dag:
    task1 = PythonOperator(
        task_id='task1',
        python_callable=task1_function
    )
    task2 = PythonOperator(
        task_id='task2',
        python_callable=task2_function
    )
    task3 = PythonOperator(
        task_id='task3',
        python_callable=task3_function
    )

    task1 >> task2 >> task3

```

After pasting the code:

Press **Enter**

Press **Ctrl + D**

This saves the file.

9. **Check file creation:** Execute: **ls**
*You should see **first_dag.py**.*
10. **Come out of container:** Execute: **exit**

Now you are back to Ubuntu terminal.

11. **Restart Airflow container:** Execute: **docker restart airflow**
Wait for 1 minute for Airflow to load the new DAG.
12. **Run the DAG in UI:**
Go back to the Airflow UI.
On the DAGs page, search for *first_dag*, click it, and then **Trigger DAG**.
Check the task status (Green means success).

Apache Airflow Docker Setup — Step by Step

1. Start Airflow container

```
docker run -d -p 8080:8080 --name airflow apache/airflow standalone
```

This starts Apache Airflow in the background.

2. Check whether container is running

```
docker ps
```

You should see a container named:

```
airflow
```

3. Check Airflow username and password

```
docker logs airflow
```

Look for the login details in the logs.

Usually:

```
username: admin
```

```
password: generated password
```

4. Go inside the Airflow container

```
docker exec -it airflow bash
```

Now you are inside the Airflow environment.

5. Move to DAGs folder

```
cd /opt/airflow/dags
```

Check location:

```
pwd
```

It should show:

```
/opt/airflow/dags
```

6. Create DAG file

```
cat > first_dag.py
```

Now paste your Python DAG code.

After pasting:

1. Press **Enter**
2. Press **Ctrl + D**

This saves the file.

7. Check whether file is created

ls

You should see:

```
first_dag.py
```

8. Come out of container

```
exit
```

Now you are back to Ubuntu terminal.

9. Restart Airflow container

```
docker restart airflow
```

Wait for 1 minute.

10. Open Airflow UI

Go to browser and open:

```
http://localhost:8080
```

Login using username and password from Docker logs.

11. Run the DAG

In Airflow UI:

DAGs page → Search first_dag → Click the DAG → Trigger DAG

Then check the task status.

Green means success 